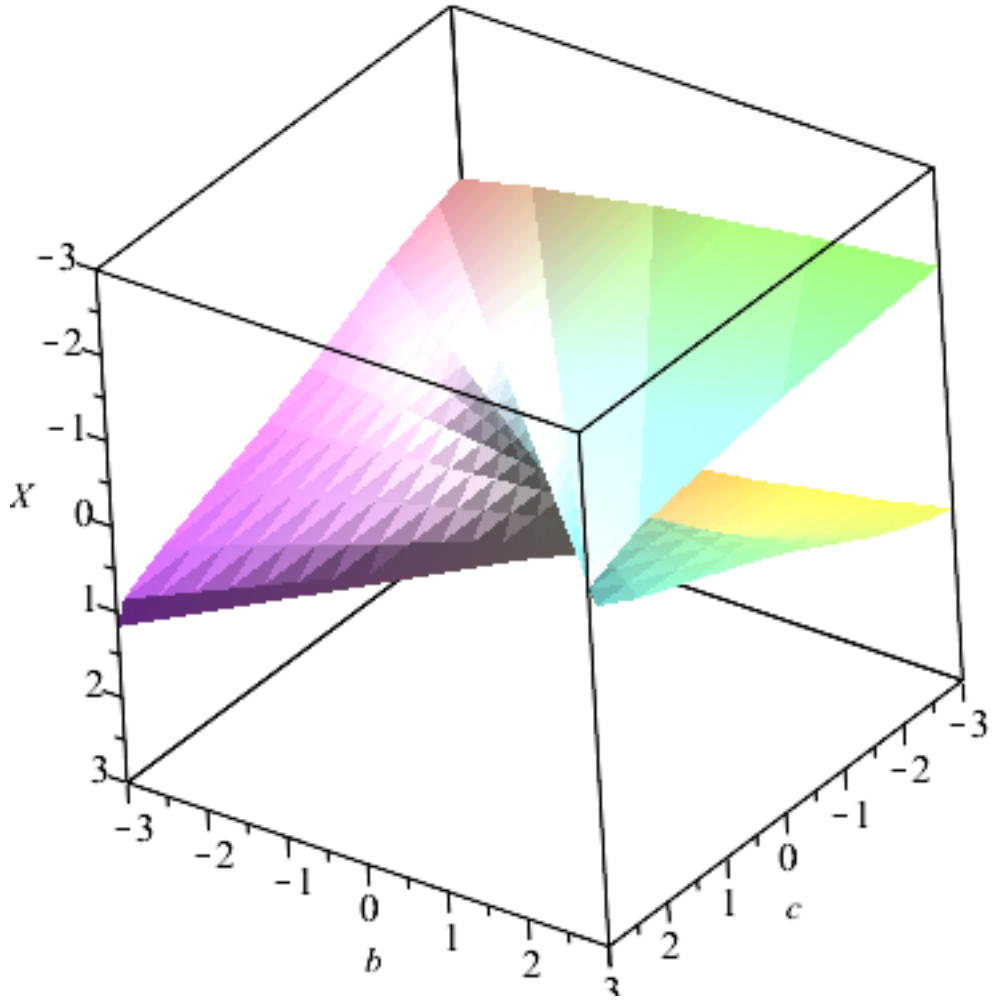
```
> restart :
>
> SW_tail := X^4 + a·X^2 + b·X + c;
```

$$SW_tail := X^4 + X^2\, a + X\, b + c \tag{1}$$

```
> SW_tail_surf1 := subs(a = -1, SW_tail);
```

$$SW_tail_surf1 := X^4 - X^2 + X\, b + c \tag{2}$$

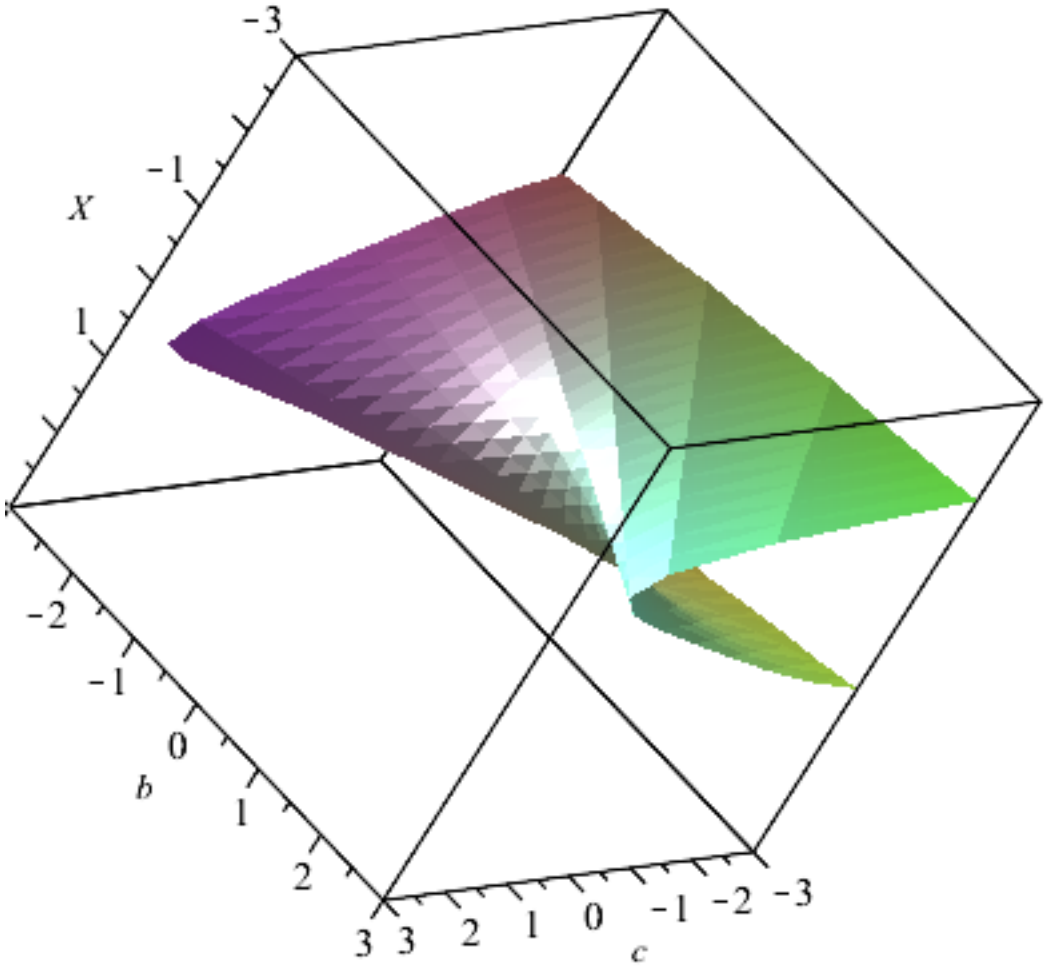
```
> with(plots) :
>
> implicitplot3d(SW_tail_surf1, X = -3 ..3, b = -3 ..3, c = -3 ..3, numpoints = 10000);
```




```
> # поверхность "ласточкин хвост" при a=-1
>
> SW_tail_surf2 := subs(a = 0, SW_tail);
```

$$SW_tail_surf2 := X^4 + X\, b + c \tag{3}$$

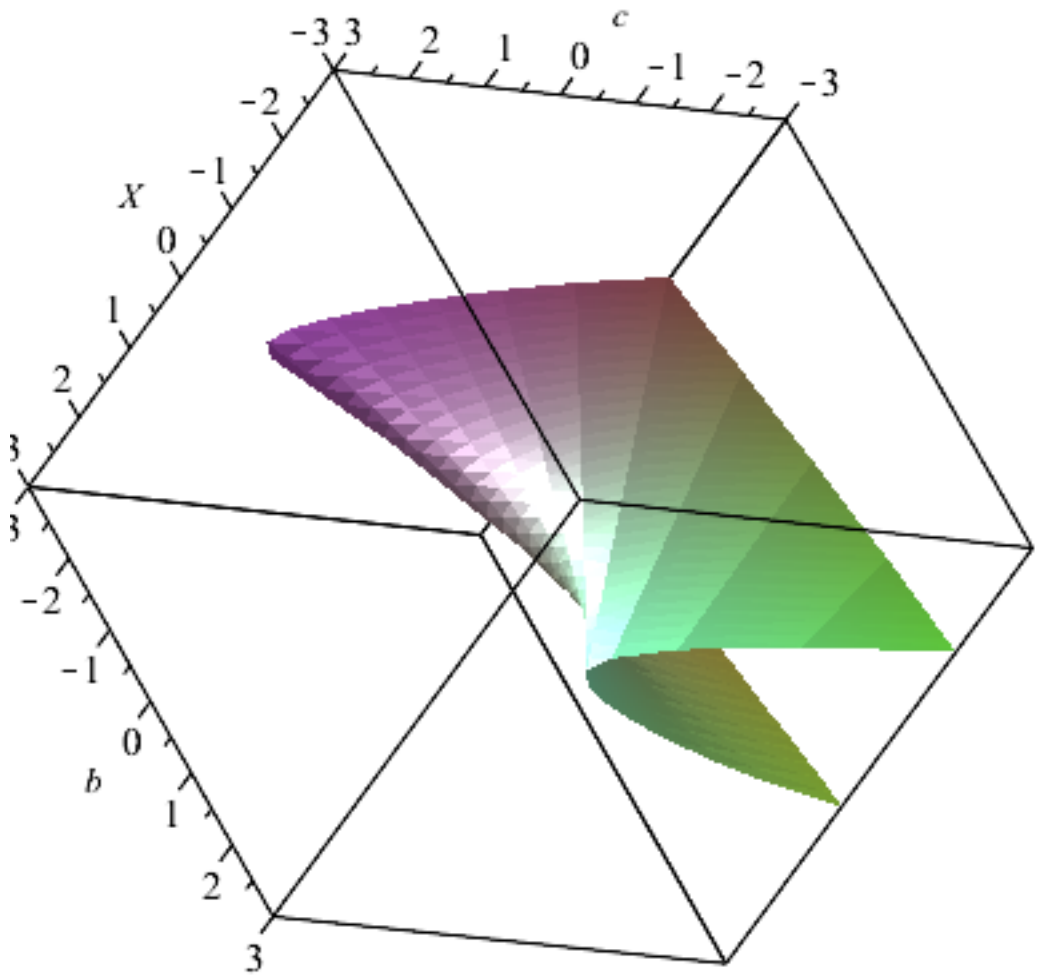
```
> implicitplot3d(SW_tail_surf2, X = -3 ..3, b = -3 ..3, c = -3 ..3, numpoints = 10000);
```




```
> # поверхность "ласточкин хвост" при a=0
>
> SW_tail_surf3 := subs(a = 1, SW_tail);
```

$$SW_tail_surf3 := X^4 + X^2 + X\,b + c \tag{4}$$

```
> implicitplot3d(SW_tail_surf3, X = -3 ..3, b = -3 ..3, c = -3 ..3, numpoints = 20000);
```




```
> # поверхность "ласточкин хвост" при a=1
>
> restart :
> with(plots) :
>
> eq1 := b = -4·X^3 - 2·a·X; eq2 := c = 3·X^4 + a·X^2; eq3 := X;
```

$$eq1 := b = -4\,X^3 - 2\,X\,a$$

$$eq2 := c = 3\,X^4 + X^2\,a$$

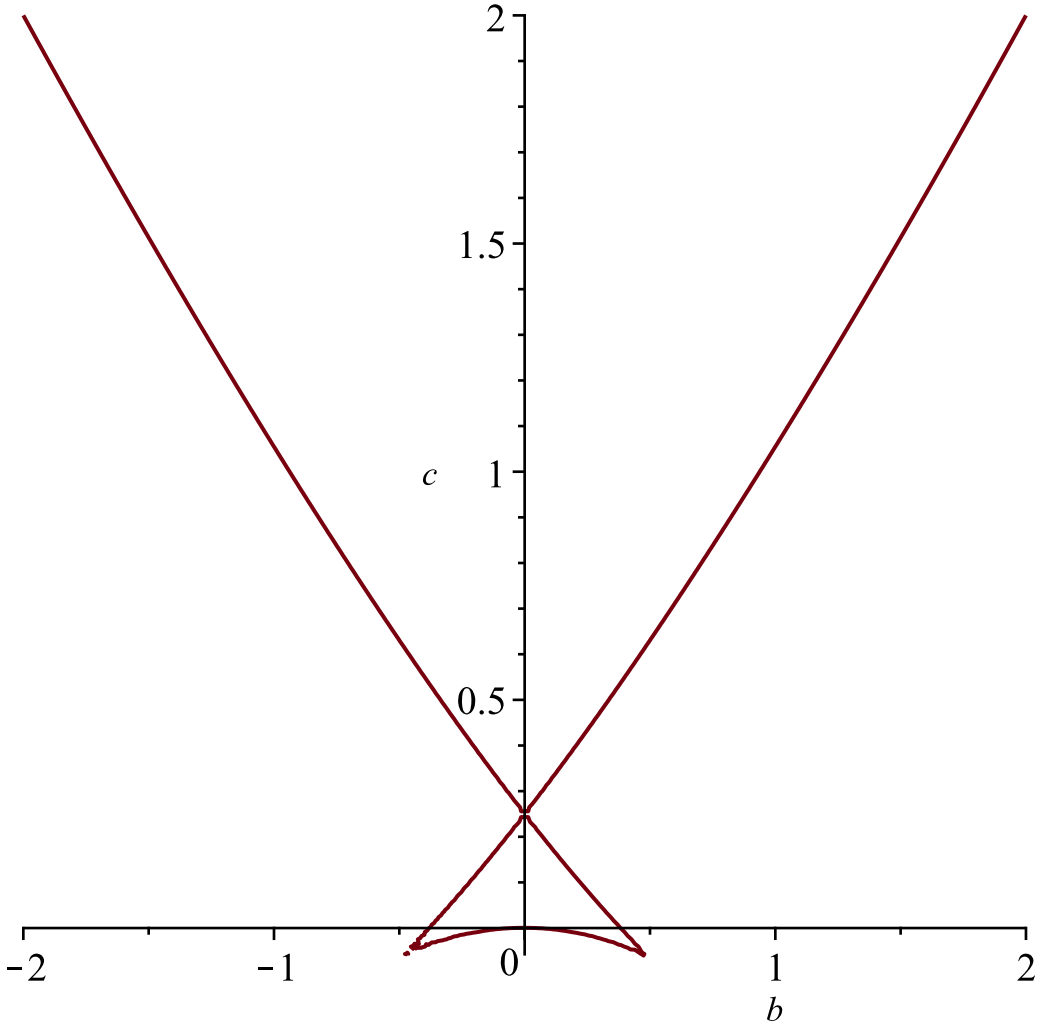
$$eq3 := X \tag{5}$$

```
> eq11 := subs(a = -1, eq1); eq12 := subs(a = -1, eq2);
```

$$eq11 := b = -4\,X^3 + 2\,X$$

$$eq12 := c = 3\,X^4 - X^2 \tag{6}$$

```
>
> implicitplot([eq11, eq12, X], b = -2..2, c = -0.5..2, numpoints = 20000);
```



> # *бифуркационная кривая "ласточкин хвост" при a=-1*

>

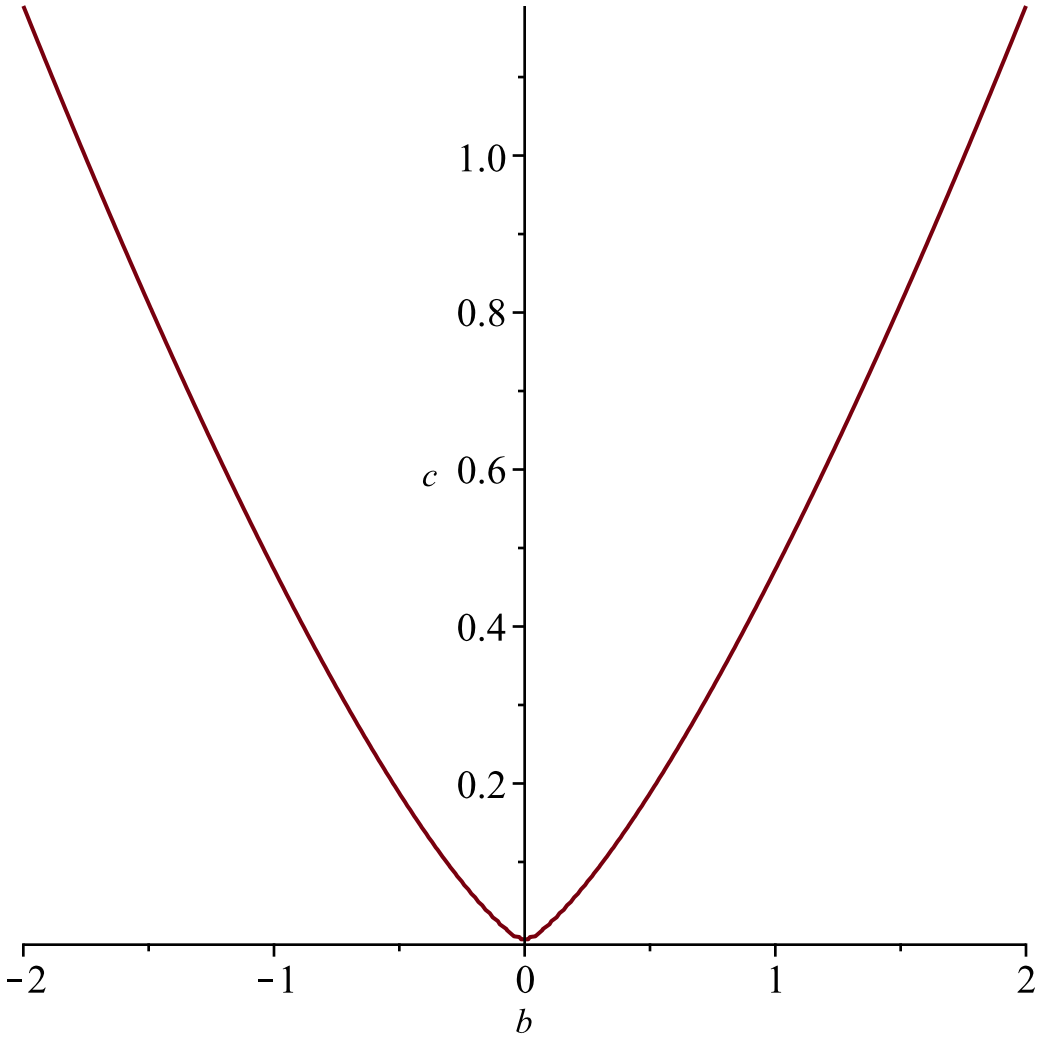
> $eq21 := subs(a = 0, eq1);\ eq22 := subs(a = 0, eq2);$

$$eq21 := b = -4\,X^3$$

$$eq22 := c = 3\,X^4 \qquad \textbf{(7)}$$

>

> $implicitplot([eq21, eq22, X], b = -2..2, c = -0.5..2, numpoints = 20000);$



> # ***бифуркационная кривая*** **"ласточкин хвост"** ***при a*** **= 0**

>

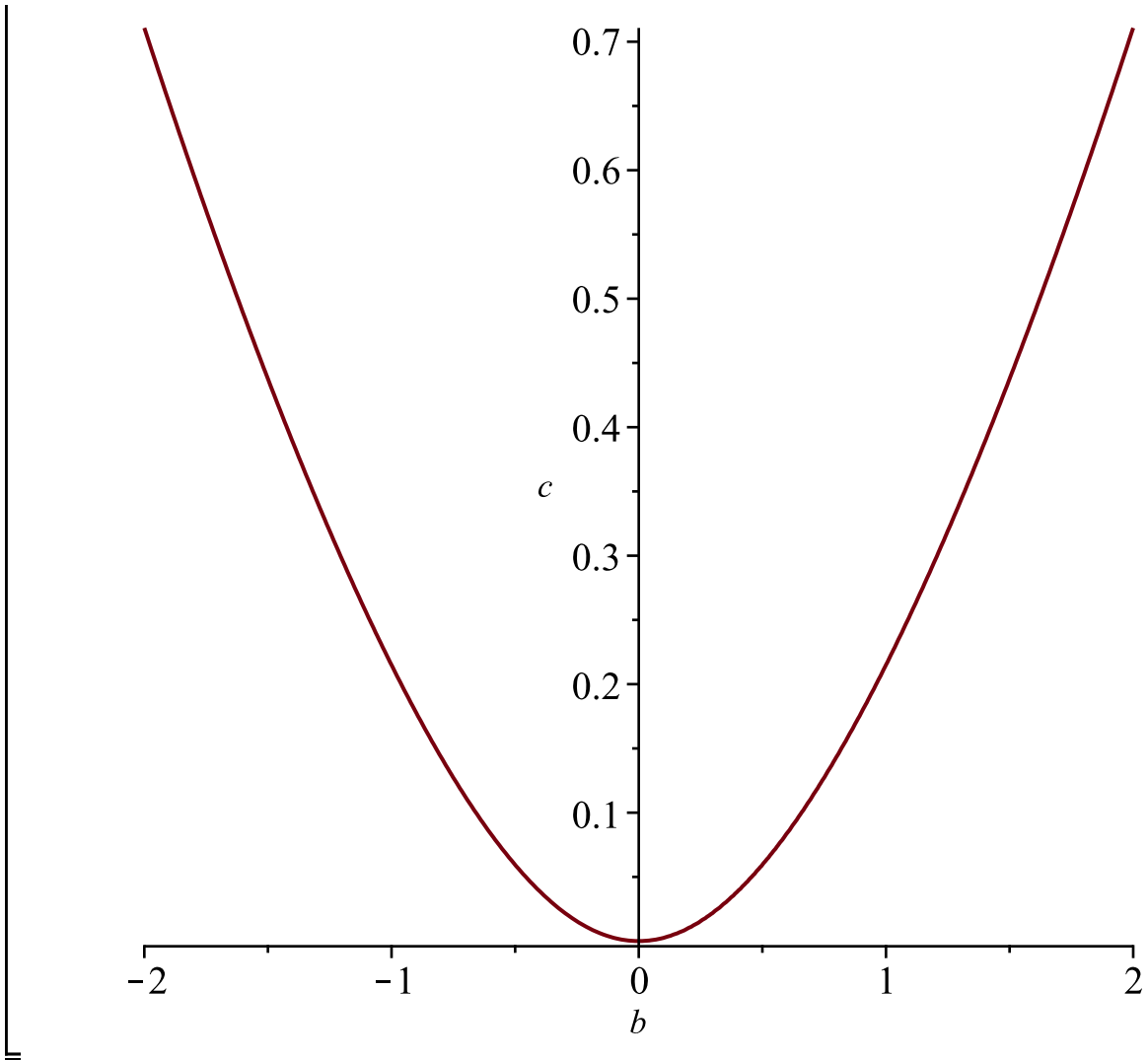
> $eq31 := subs(a = 1, eq1);\ eq32 := subs(a = 1, eq2);$

$$eq31 := b = -4X^3 - 2X$$

$$eq32 := c = 3X^4 + X^2 \qquad \textbf{(8)}$$

>

> $implicitplot([eq31, eq32, X], b = -2..2, c = -0.5..2, numpoints = 20000);$



> # ***бифуркационная кривая*** "ласточкин хвост" ***при a*** **= 1**

```
> restart :
>
> # КАТАСТРОФА БАБОЧКА
>
> with(plots) :
>
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (1)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (2)$$

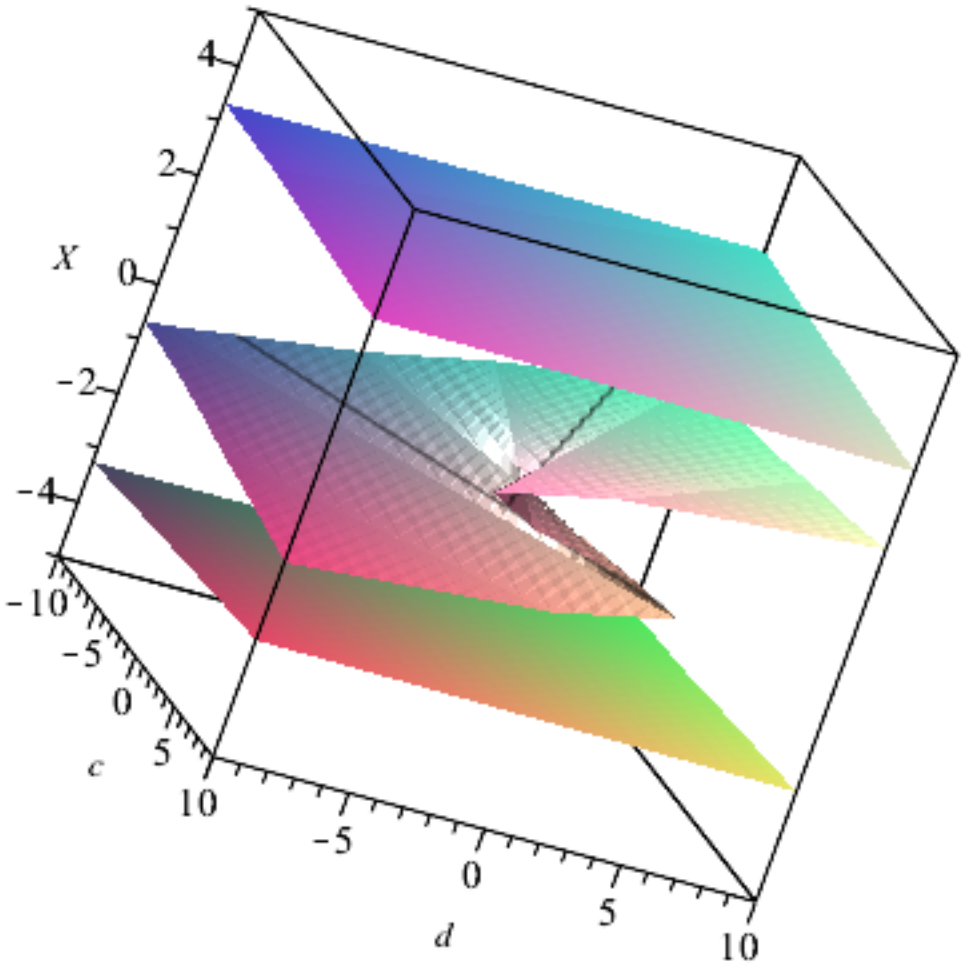
```
>
> eq3 := subs(a = -10, b = 0, eq1);
```

$$eq3 := X^5 - 10\,X^3 + X\,c + d = 0 \quad (3)$$

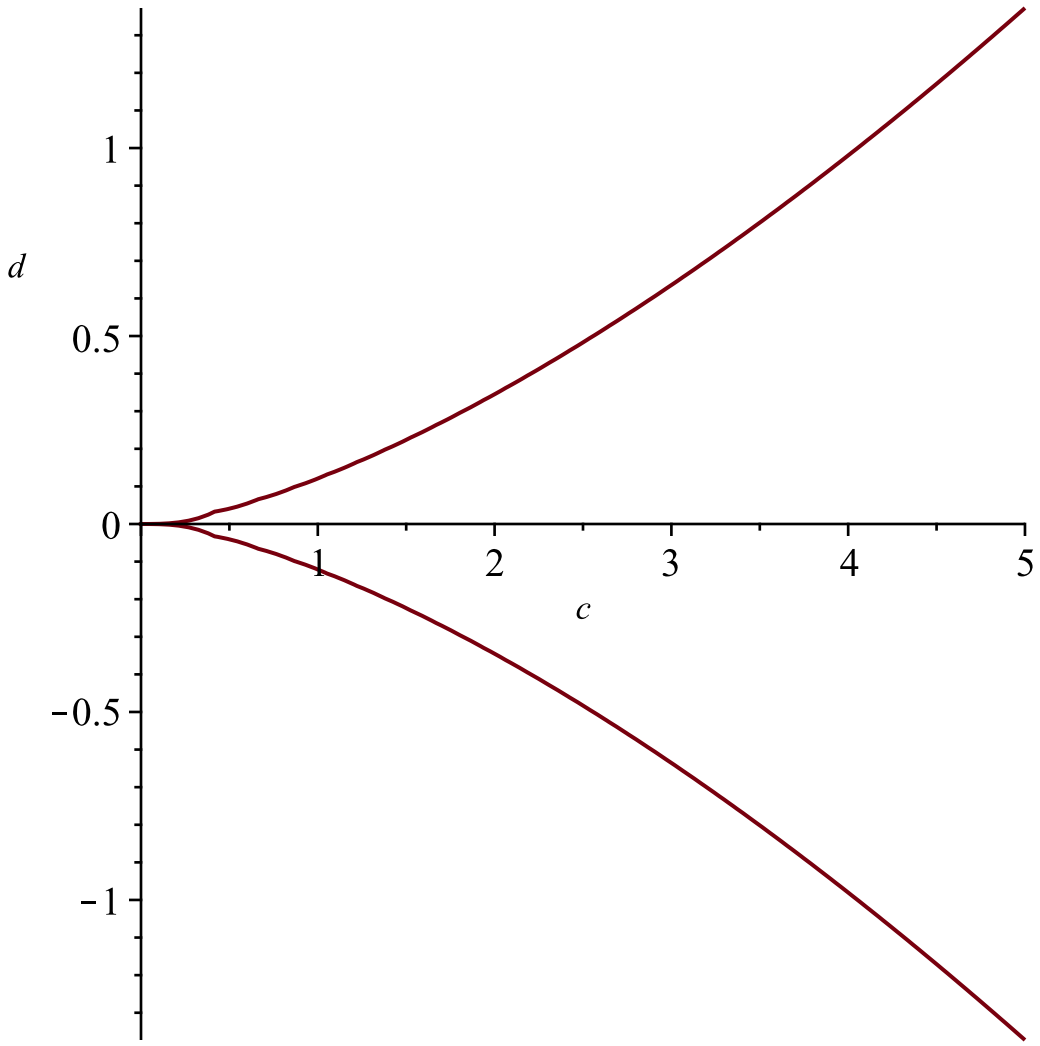
```
> eq4 := subs(a = -10, b = 0, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 + c = 0 \quad (4)$$

```
> implicitplot3d(eq3, c = -10 ..10, d = -10 ..10, X = -5 ..5, numpoints = 10000);
```



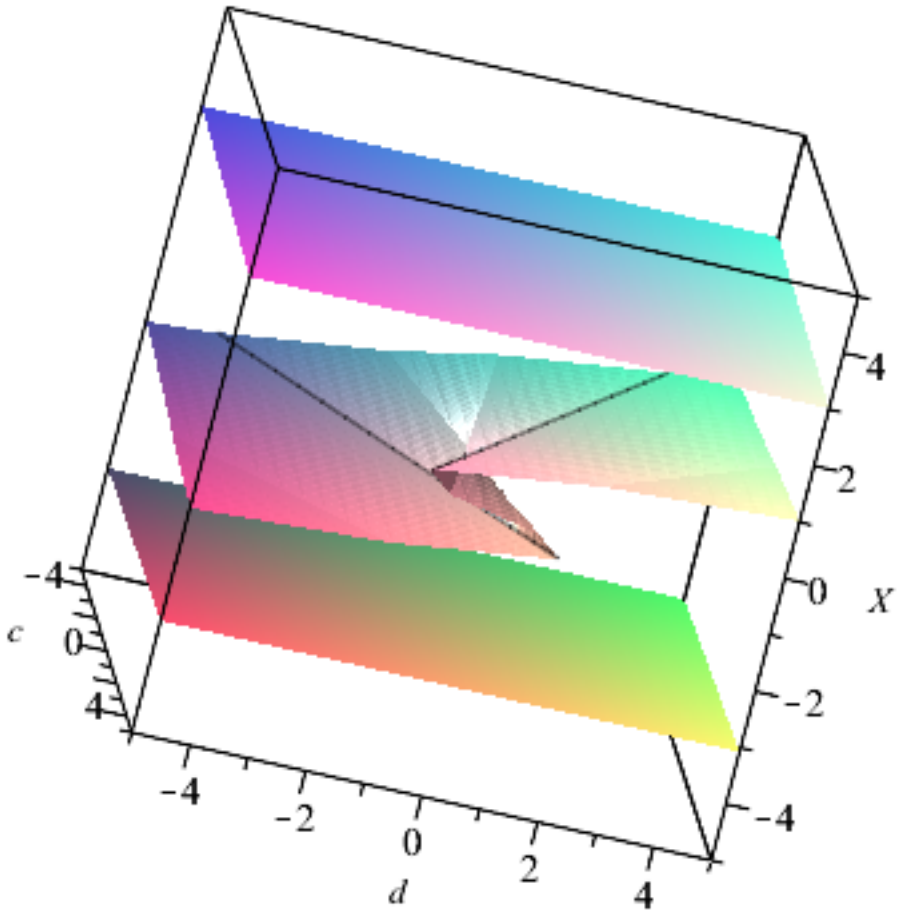

```
> implicitplot([eq3, eq4, X], c = -5 ..5, d = -5 ..5, numpoints = 10000);
```




```
>
> surf_butterfly2 := subs(a = -6, b = 10, eq3);
```

$$surf_butterfly2 := X^5 - 10\,X^3 + X\,c + d = 0 \quad \textbf{(5)}$$

```
> implicitplot3d(eq3, c = -5..5, d = -5..5, X = -5..5, numpoints = 20000);
```




```
> restart :
>
> with(plots) :
>
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (6)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (7)$$

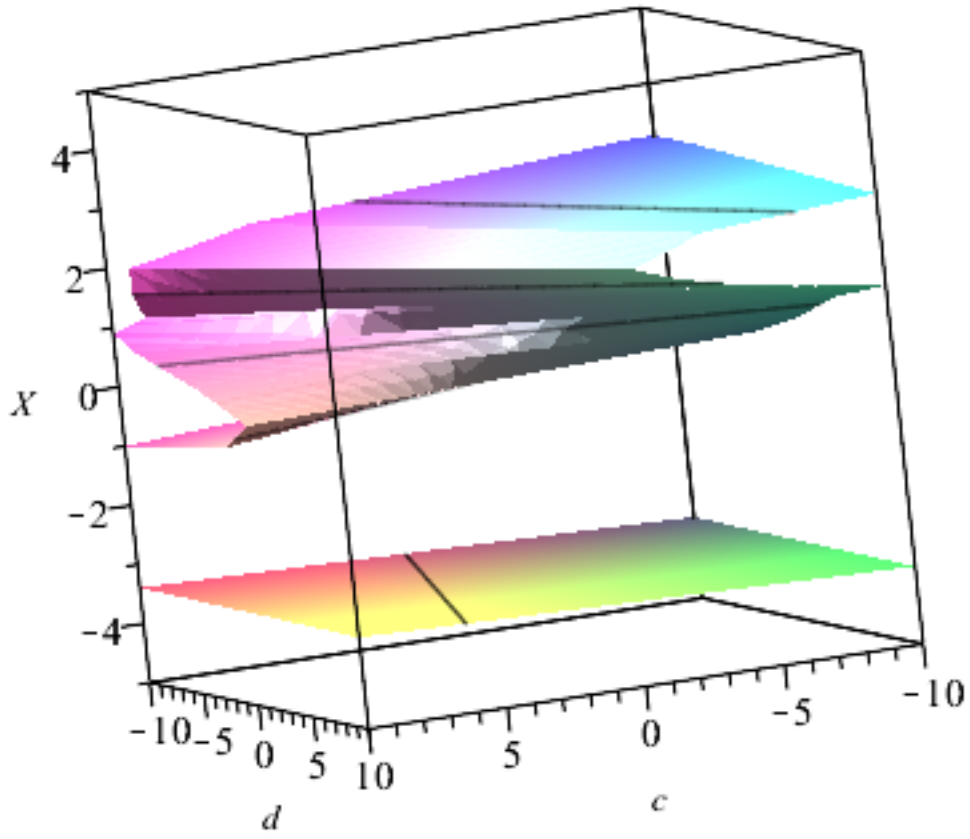
```
>
> eq3 := subs(a = -10, b = 10, eq1);
```

$$eq3 := X^5 - 10\,X^3 + 10\,X^2 + X\,c + d = 0 \quad (8)$$

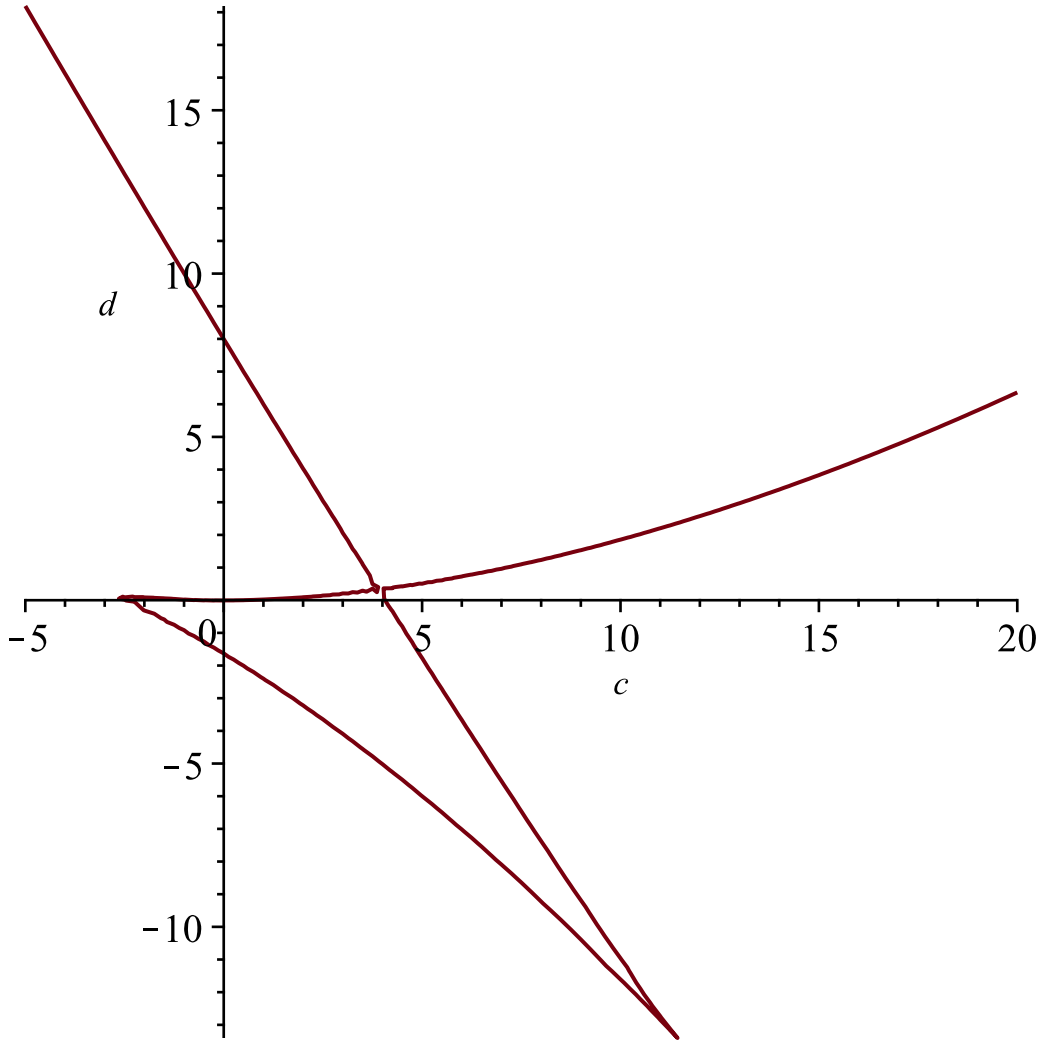
```
> eq4 := subs(a = -10, b = 10, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 + 20\,X + c = 0 \quad (9)$$

```
> implicitplot3d(eq3, c = -10..10, d = -10..10, X = -5..5, numpoints = 10000);
```



> *implicitplot*([*eq3*, *eq4*, *X*], *c* = −5 ..20, *d* = −20 ..20, *numpoints* = 10000);




```
> restart :
> with(plots) :
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (10)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (11)$$

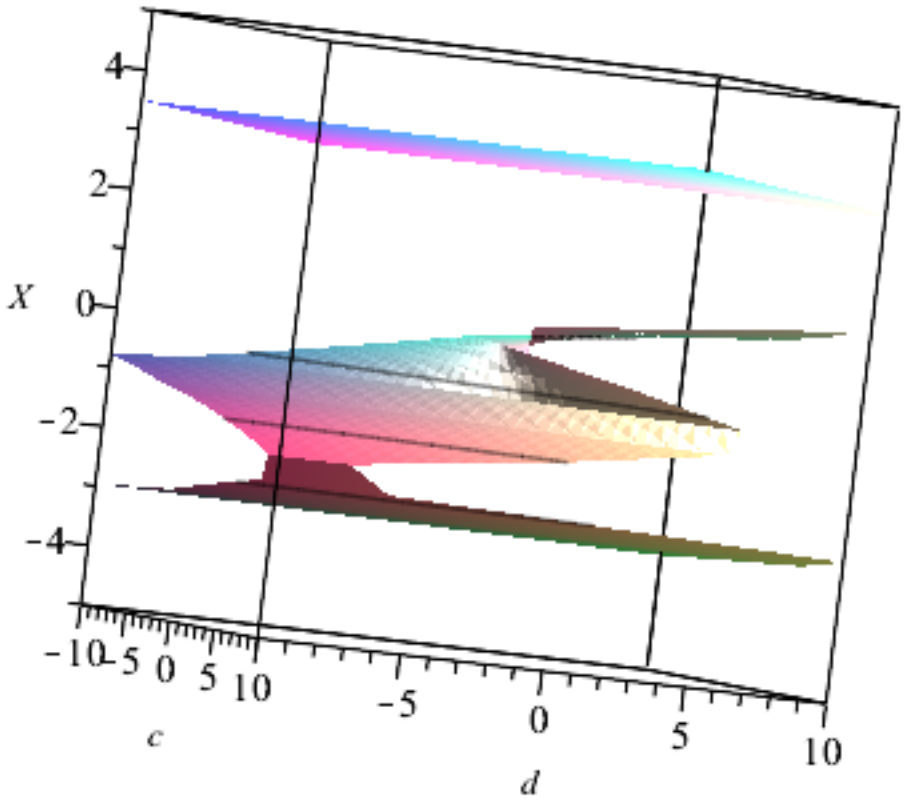
```
> eq3 := subs(a = -10, b = -5, eq1);
```

$$eq3 := X^5 - 10\,X^3 - 5\,X^2 + X\,c + d = 0 \quad (12)$$

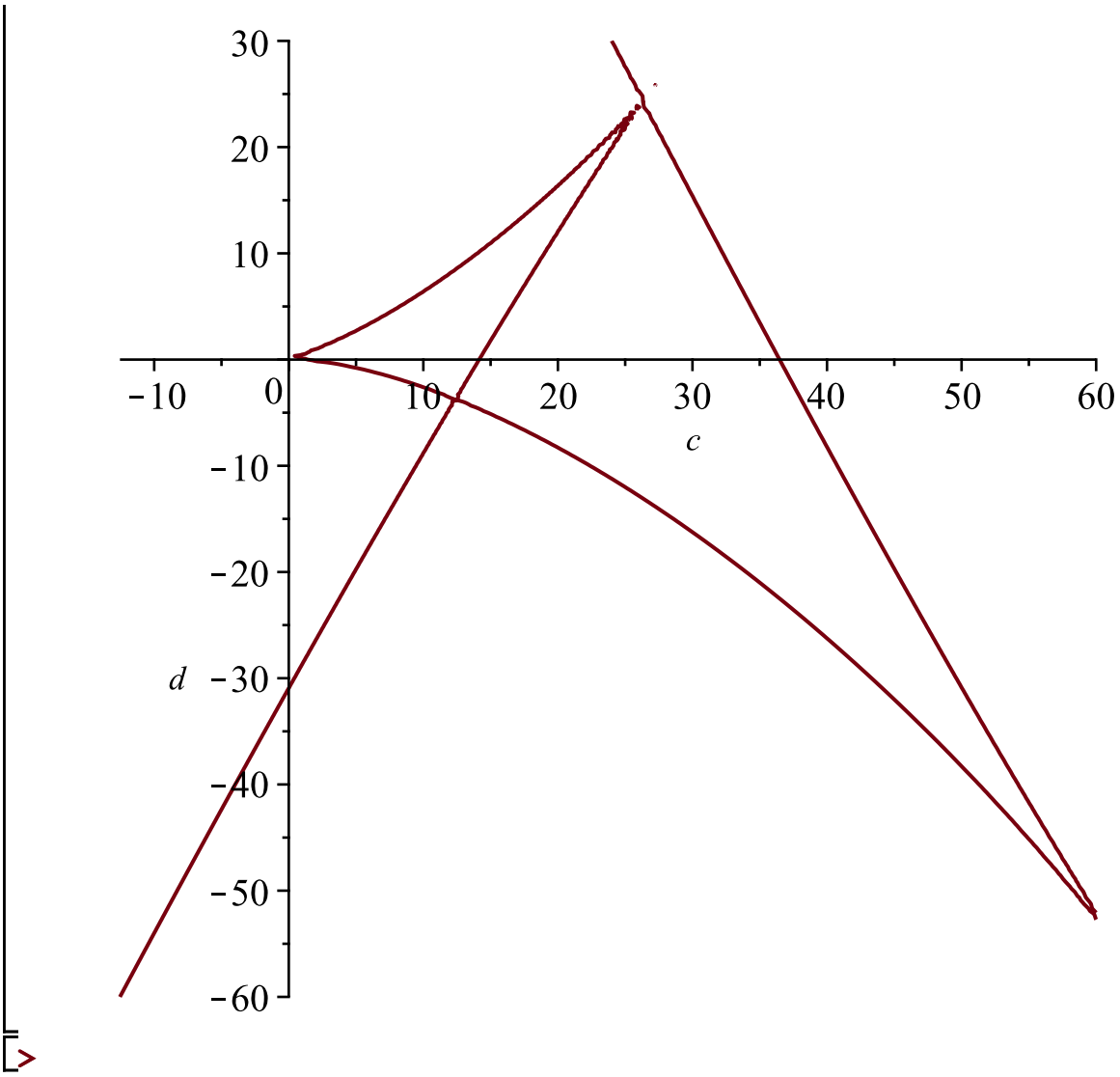
```
> eq4 := subs(a = -10, b = -5, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 - 10\,X + c = 0 \quad (13)$$

```
> implicitplot3d(eq3, c = -10..10, d = -10..10, X = -5..5, numpoints = 10000);
```




```
> implicitplot( [eq3, eq4, X], c =-20 ..60, d =-60 ..30, numpoints = 30000 );
```


30
20
10
-10
0
10
20
30
40
50
60
c
-10
-20
d
-30
-40
-50
-60